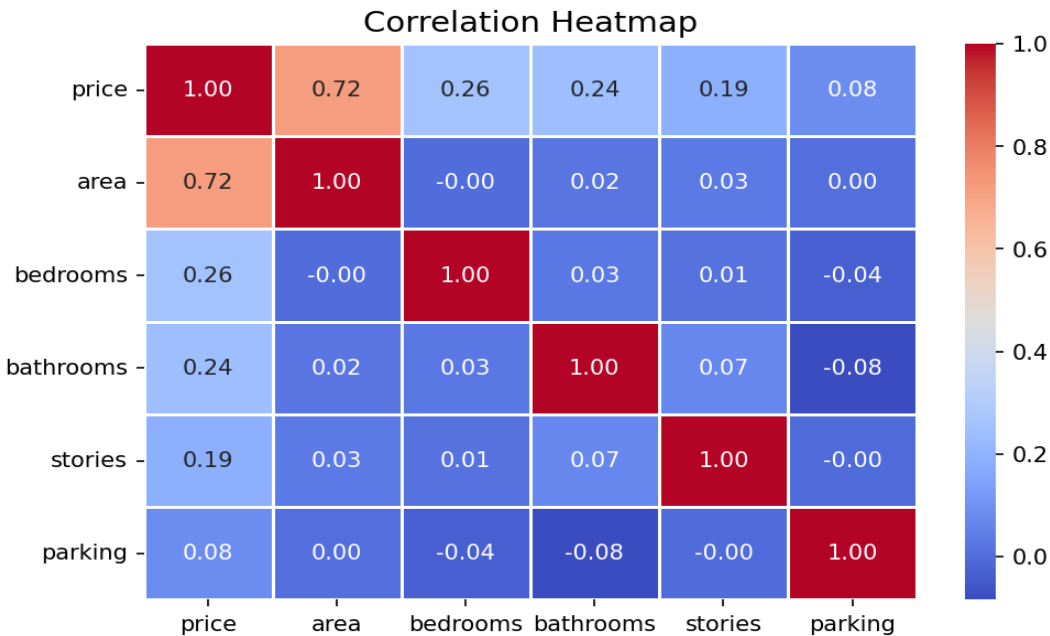


House Price Prediction — Findings Summary

Which Features Influence House Price the Most?

Based on the correlation heatmap and model analysis, **area (sq. ft.)** is the strongest driver of house price. The number of **bathrooms** and **bedrooms** rank next, followed by **stories** and **parking**. Location attributes such as being in a **preferred area** and having **main road access** also add meaningful value, while amenities like air conditioning and basement presence contribute at a smaller scale.



How Accurate Was the Model? (Plain Terms)

Linear Regression explained **67.2%** of house price variation ($R^2 = 0.6722$), a reasonable result for real housing data. **Random Forest** achieved 63.0% ($R^2 = 0.6305$), slightly lower — confirming that price relationships here are largely linear.

Model	R ² Score	MAE	RMSE
Linear Regression	0.6722	0.1999	0.2516
Random Forest	0.6305	0.2060	0.2671

What Surprised Us in the Data?

Both models achieved similar R^2 scores (~63–67%), indicating house prices are influenced by factors beyond the available features such as neighbourhood sentiment or market timing. It was also surprising that **Linear Regression outperformed Random Forest**, and that the dataset had **zero missing values and zero duplicates** — rare in real-world data.

Business Recommendation

Prioritize **larger properties in preferred locations with main road access**, as area and location are the dominant price drivers. To improve model accuracy beyond 67%, enrich the dataset with features like proximity to schools, market trends, or neighbourhood ratings before deploying any automated valuation tool.